

AEROFLEX-ANDOVER, NJ, USA

WMO: 724077

Lat: 41.009N

Lon: 74.736W

Elev: 580

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54779

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.5	(c) 9.4	(d) -8.9	(e) 3.5	(f) 12.6	(g) -3.9	(h) 4.6	(i) 13.4	(j) 15.9	(k) 31.4	(l) 13.9	(m) 31.5	(n) 2.9	(o) 0	(p) 0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.7	(c) 89.5	(d) 73.4	(e) 86.6	(f) 72.2	(g) 84.0	(h) 70.7	(i) 76.2	(j) 84.6	(k) 74.7	(l) 82.4	(m) 73.3	(n) 80.3	(o) 5.4	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	128.0	79.8	72.4	122.5	78.4	71.2	117.5	77.1	40.0	84.8	38.5	82.8	37.3	80.6	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.8	(b) 11.3	(c) 10.2	(e) -2.4	(f) 93.8	(g) 7.2	(h) 2.9	(i) -7.6	(j) 95.9	(k) -11.8	(l) 97.6	(m) -15.8	(n) 99.2	(o) -21.0	(p) 101.3		
(4) 12.8	(5) 11.3	(6) 10.2	(7) -2.4	(8) 93.8	(9) 7.2	(10) 2.9	(11) -7.6	(12) 95.9	(13) -11.8	(14) 97.6	(15) -15.8	(16) 99.2	(17) -21.0	(18) 101.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	27.9	29.4	38.1	50.5	60.3	68.3	73.0	71.3	64.6	53.3	42.7	34.1
	DBStd	17.63	10.05	9.33	9.53	8.63	7.72	6.03	4.85	4.93	6.37	8.10	8.20	8.61
	HDD50	2550	687	578	385	94	7	0	0	0	58	245	496	
	HDD65	5737	1151	996	834	440	190	35	4	7	85	370	669	956
	CDD50	3008	1	2	16	110	325	549	714	661	437	162	27	4
	CDD65	719	0	0	0	6	44	134	252	203	71	9	0	0
	CDH74	5937	0	0	2	132	456	1124	2086	1527	552	57	1	0
	CDH80	1744	0	0	0	35	120	311	696	440	135	7	0	0
Wind	WSAvg	3.4	4.0	4.1	4.5	4.4	3.3	2.8	2.5	2.2	2.5	3.3	3.7	3.8
Precipitation	PrecAvg	45.0	3.4	2.6	3.8	3.5	3.5	4.5	4.0	4.3	4.2	3.9	3.3	3.9
	PrecMax	67.2	7.8	7.1	8.7	7.6	7.8	9.8	7.3	13.0	11.3	14.5	5.1	7.9
	PrecMin	27.0	1.6	0.7	0.8	1.7	0.9	0.9	1.1	1.3	0.1	0.0	1.3	1.1
	PrecStd	9.5	1.5	1.3	1.9	1.5	1.6	2.3	1.5	2.4	3.1	3.2	1.1	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.3	62.6	72.1	84.6	87.5	91.2	93.1	92.4	89.0	79.6	69.9	63.1
		MCWB	56.0	55.4	57.5	62.8	69.9	74.1	74.9	74.3	72.7	66.9	59.9	58.3
	2%	DB	53.3	53.7	64.3	77.3	83.6	86.6	89.6	87.9	83.9	74.5	64.5	56.4
		MCWB	48.2	45.7	52.3	58.4	67.7	71.8	74.2	74.0	70.4	64.1	57.6	52.4
	5%	DB	46.3	47.5	58.0	72.1	79.4	83.3	86.6	84.4	80.3	70.9	60.7	51.4
		MCWB	41.3	41.0	48.0	57.7	64.9	70.2	72.6	71.7	68.1	63.0	54.0	47.0
	10%	DB	41.9	42.5	53.2	66.0	74.8	80.8	83.9	81.9	76.7	66.9	56.4	46.9
		MCWB	37.6	36.9	45.3	53.3	62.8	68.7	71.3	70.2	66.9	60.5	50.3	41.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	57.1	59.5	67.1	72.9	76.2	78.3	78.3	76.2	70.7	63.5	59.5
		MCDB	59.9	62.5	67.4	77.4	83.4	86.4	88.5	86.7	84.1	75.1	67.4	62.1
	2%	WB	49.5	47.5	54.6	62.8	70.1	74.2	76.3	76.0	73.2	67.6	59.3	53.5
		MCDB	51.7	52.9	62.1	72.7	79.6	83.1	85.1	83.3	79.0	72.0	63.2	55.8
	5%	WB	42.4	41.3	50.1	59.4	68.0	72.5	74.9	74.3	71.2	64.7	55.5	47.9
		MCDB	45.7	46.1	56.2	68.8	75.5	80.5	82.8	81.0	76.8	69.5	59.5	50.6
	10%	WB	38.1	37.3	46.0	55.7	65.7	70.7	73.4	72.6	69.3	61.1	51.6	42.5
		MCDB	41.3	42.0	52.5	63.8	72.1	77.8	80.5	78.8	74.0	65.8	55.0	45.8
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.0	18.9	21.0	20.7	19.7	19.7	19.6	20.5	19.0	17.4	14.5
		MCDBR	23.6	25.4	26.4	29.6	26.1	23.0	22.5	22.0	24.3	23.7	23.3	19.6
		MCWBR	19.4	18.5	16.4	14.1	12.1	10.7	9.4	9.6	12.0	13.5	15.9	16.6
	5% WB	MCDBR	21.1	22.8	22.9	24.7	21.3	20.2	19.0	18.6	19.2	19.3	19.2	18.4
		MCWBR	19.0	17.9	17.1	14.6	11.1	10.4	8.9	9.0	10.7	13.3	16.0	17.9
Clear-Sky Solar Irradiance	taub	0.299	0.318	0.349	0.370	0.413	0.473	0.474	0.461	0.380	0.354	0.319	0.299	
	taud	2.444	2.388	2.387	2.368	2.273	2.140	2.150	2.176	2.380	2.463	2.518	2.514	
	Ebn at Noon	273	283	285	286	275	257	256	255	272	267	264	264	
	Edh at Noon	25	30	34	37	42	48	47	45	34	28	23	22	
All-Sky Solar Radiation	RadAvg	569	851	1147	1474	1643	1791	1850	1597	1335	915	638	475	
	RadStd	46	90	93	149	175	163	98	104	152	100	61	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	+115

Nomenclature: See separate page